Applications of Integrals

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Contents

1 Area Under a Curve **2**

2 Area with Respect to the y-Axis **2**

3 Area Between Two Curves **2**

4 Standard Curves and Symmetry **2**

5 Areas of a Circle and an Ellipse **3**

6 Steps for Area Problems **3**

1 Area Under a Curve

For a continuous function $y = f(x)$ with $f(x) \geq 0$ on $[a, b]$, the area between the curve, the x-axis, and $x = a$, $x = b$ is

$$\boxed{A = \int_a^b f(x)\,dx.}$$

If $f(x) \leq 0$, the geometric area is $-\int_a^b f(x)\,dx$. When the graph crosses the axis, split the interval at the zeros and add the positive areas:

$$A = \int_a^b |f(x)|\,dx.$$

Example. Area under $y = x^2$ from $x = 0$ to $x = 2$:

$$A = \int_0^2 x^2\,dx = \left[\frac{x^3}{3}\right]_0^2 = \frac{8}{3}.$$

2 Area with Respect to the y-Axis

For $x = g(y) \geq 0$ on $[c, d]$, the area bounded by the curve, the y-axis, and horizontal lines $y = c, d$ is

$$\boxed{A = \int_c^d g(y)\,dy.}$$

Choose horizontal strips when x is easier to express in terms of y.

3 Area Between Two Curves

If $f(x) \geq g(x)$ on $[a, b]$, the enclosed area is

$$\boxed{A = \int_a^b \big(f(x) - g(x)\big)\,dx.}$$

Find intersection points by solving $f(x) = g(x)$. If the upper curve changes, split the integral at intersections. **Example.** Between $y = x$ and $y = x^2$:

$$x = x^2 \implies x = 0, 1, \qquad A = \int_0^1 (x - x^2)\,dx = \frac{1}{2} - \frac{1}{3} = \boxed{\frac{1}{6}}.$$

For horizontal strips, use $A = \int_c^d (x_{\text{right}} - x_{\text{left}})\,dy$.

4 Standard Curves and Symmetry

- Lines: $y = mx + c$.
- Parabolas: $y^2 = 4ax$, $x^2 = 4ay$, and their reflected forms.
- Circles: $x^2 + y^2 = r^2$, with $y = \pm\sqrt{r^2 - x^2}$.
- Ellipses: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} = 1$, with $y = \pm b\sqrt{1 - x^2/a^2}$.

For a region symmetric about an axis, compute one symmetric portion and multiply by 2 (or by 4 for four identical quadrants).

5 Areas of a Circle and an Ellipse

For a circle $x^2 + y^2 = r^2$, symmetry gives

$$A = 4\int_0^r \sqrt{r^2 - x^2}\,dx = \boxed{\pi r^2}.$$

For an ellipse $x^2/a^2 + y^2/b^2 = 1$,

$$A = 4\int_0^a b\sqrt{1 - \frac{x^2}{a^2}}\,dx = \boxed{\pi ab}.$$

6 Steps for Area Problems

1. Sketch the curves and mark the bounded region.
2. Find intersections and the relevant limits.
3. Decide whether vertical or horizontal strips are simpler.
4. Set up upper minus lower (or right minus left).
5. Split the integral if the ordering changes; evaluate and report square units.